

structure works well in situations in which you want Python to always execute one of two possible actions. In a simple if-else chain like this, one of the two actions will always be executed.

The if-elif-else Chain

Often, you'll need to test more than two possible situations, and to evaluate these you can use Python's if-elif-else syntax. Python executes only one block in an if-elif-else chain. It runs each conditional test in order, until one passes. When a test passes, the code following that test is executed and Python skips the rest of the tests.

Many real-world situations involve more than two possible conditions. For example, consider an amusement park that charges different rates for different age groups:

- Admission for anyone under age 4 is free.
- Admission for anyone between the ages of 4 and 18 is \$25.
- Admission for anyone age 18 or older is \$40.

How can we use an if statement to determine a person's admission rate? The following code tests for the age group of a person and then prints an admission price message:

```
amusement    age = 12
_park.py ❶ if age < 4:
            print("Your admission cost is $0.")
          ❷ elif age < 18:
            print("Your admission cost is $25.")
          ❸ else:
            print("Your admission cost is $40.")
```

The if test ❶ checks whether a person is under 4 years old. When the test passes, an appropriate message is printed and Python skips the rest of the tests. The elif line ❷ is really another if test, which runs only if the previous test failed. At this point in the chain, we know the person is at least 4 years old because the first test failed. If the person is under 18, an appropriate message is printed and Python skips the else block. If both the if and elif tests fail, Python runs the code in the else block ❸.

In this example the if test ❶ evaluates to False, so its code block is not executed. However, the elif test evaluates to True (12 is less than 18) so its code is executed. The output is one sentence, informing the user of the admission cost:

```
Your admission cost is $25.
```

Any age greater than 17 would cause the first two tests to fail. In these situations, the else block would be executed and the admission price would be \$40.

Rather than printing the admission price within the if-elif-else block, it would be more concise to set just the price inside the if-elif-else chain